

Economic Basics

1. Needs, Wants and Desires

Need

A **need** is something essential for survival and well-being.

Examples:

- Food
- Clothing
- Shelter
- Healthcare

The key idea is:

Needs are basic requirements necessary for survival and well-being.

Want

Once our basic needs are fulfilled, our aspirations increase and we start looking for things that improve our **quality of life**.

Examples:

- Entertainment
- Travel
- Other goods and services beyond basic necessities

So:

Needs → fulfilled → Wants emerge

Desire

The lecture distinguishes a desire from a want mainly through **affordability**.

For example:

You want to purchase something, but you don't have enough money to purchase it.

That desired item therefore represents something you wish to have but cannot currently afford.

Concept	Meaning	Example
Need	Essential for survival/well-being	Food
Want	Desired to improve quality of life	Vacation
Desire	Something you wish to have but may not be able to afford	Expensive car

2. Infinite / Insatiable Wants

One of the **most important ideas in economics** is that human wants are considered **infinite** or **insatiable**.

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This means:

Humans are generally not satisfied with what they already have and tend to want more.

Important distinction

Infinite wants ≠ selfishness

The lecturer specifically points out that having unlimited wants does not necessarily mean humans are selfish.

Humans can simultaneously:

- Want more for themselves
- Care about other people
- Want other people to have more

Exam point

Infinite/insatiable wants: Human wants are unlimited because satisfaction of one want can lead to the emergence of further wants.

3. What Does "Wants" Mean in Economics?

In this context, wants refer to the **goods and services that individuals desire**.

Examples include:

- Food
- Clothing
- Shelter
- Other goods/services that enhance quality of life

4. Scarcity of Resources

Now we reach the central problem of economics.

Humans have:

Unlimited wants

but the resources required to satisfy those wants are:

Scarce / limited

The lecture describes scarce resources as resources that are **short in supply**.

This conflict between unlimited wants and scarce resources is the foundation of economics.

5. Constraints

Because resources are scarce, there are **constraints**.

Constraint = limitation

If resources are limited, there are limits on:

- What can be produced
- How much can be produced
- Which human wants can ultimately be satisfied

This leads directly to the definition of economics.

6. Definition of Economics

The lecture gives two closely related definitions.

Definition 1

Economics is the science of constrained choices.

Definition 2

Economics deals with the allocation of scarce resources to meet unlimited human wants.

7. Types of Goods

Economists classify goods into two broad categories in this lecture:

1. **Economic goods**
2. **Free goods**

7.1 Economic Goods

Economic goods are goods that are **scarce in nature**.

Their available resources/supply are limited relative to wants.

Examples can include:

- Food
- Cars
- Buildings
- Healthcare resources

7.2 Free Goods

Free goods are goods where there is **enough supply to meet human wants**.

The lecture uses **air** as an example.

Very important clarification

Free does NOT simply mean "the consumer doesn't pay money."

This is a potential exam trap.

The lecturer gives the example of government-funded healthcare for poor people.

Even if the individual does not pay for the healthcare, it does **not** automatically make healthcare a free good.

Remember

Free good ≠ free of monetary price

Instead, the important economic question is whether **scarce resources are involved**.

8. Resources / Factors of Production

To produce goods and services, we need **resources**.

The lecture gives examples such as:

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- Land
- Labour
- Raw materials

Resources come from the Earth's available resources, but these resources are scarce.

Economists divide these scarce resources into **four factors of production**:

Factor	Meaning	Examples
Land	Natural resources	Oil, sea, land
Labour	Human effort/people available to work	Workers
Capital	Goods used to produce other goods	Buildings, machinery
Enterprise	Entrepreneurship that combines resources	Entrepreneurs

9. Land

In economics, **land** refers to scarce natural resources.

Examples mentioned:

- Oil
- Sea
- Other natural resources

Don't think of "land" as merely a physical piece of property.

In economics, the concept is broader: **natural resources used in production**.

10. Labour

Labour refers to people who are available to work and contribute their:

- Knowledge
- Intelligence
- Skills
- Human effort

to production.

11. Capital

Capital means **goods used to produce other goods**.

Examples:

- Buildings
- Machinery

Important

In this context:

Capital ≠ simply money

Economic capital refers to productive resources such as machinery and buildings.

12. Enterprise

Enterprise refers to the entrepreneurial function.

Entrepreneurs combine:

Land + Labour + Capital

to produce goods and services.

13. Production

The lecture first introduces the familiar idea:

Raw material → finished product

But economically, production is broader.

Economic definition

Production is any activity carried out that satisfies a human want.

So production doesn't necessarily mean physically manufacturing an object.

Services can also be produced.

14. The Economic Problem

This is probably the **most important concept in the entire lecture**.

We have:

Human wants = Unlimited

but:

Resources = Limited / scarce

Therefore:

We cannot have everything we want.

Consequently, resources must be **carefully allocated**.

This is called the:

Economic Problem

In one line:

Unlimited wants + scarce resources → choice/allocation → economic problem

15. Why Choice Is Necessary

Suppose a government has a limited budget.

It wants to spend money on:

- Education
- Healthcare
- Infrastructure
- Defence

But its resources are limited.

It cannot necessarily maximize spending on everything simultaneously.

Therefore it has to make a **choice**.

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This is the basic economic problem.

The same logic applies to:

- Individuals
- Businesses
- Governments
- Entire economies

Microeconomics vs Macroeconomics

Economics is divided into two major branches:

1. *Microeconomics*
2. *Macroeconomics*

Microeconomics

Microeconomics focuses on **smaller parts of the economy**.

Think:

Individuals + consumers + businesses + workers

Example from the lecture:

What is the impact of tariffs on:

- Consumers
- Individuals
- Businesses
- Workers

This is a **micro-economic perspective**.

Think:

Micro = small / individual units

18. Macroeconomics

Macroeconomics focuses on the **larger economy / entire economy**.

Examples:

- GDP
- Employment
- Economy-wide effects
- Government budget decisions

The lecture uses tariffs and their impact on India's employment rate and GDP as an example.

Think:

Macro = large / entire economy

Business Data Management (Part 1)

Microeconomics	Macroeconomics
Individual units	Entire economy
Consumers	National economy
Individual businesses	GDP
Workers	Employment
Consumer behaviour	Economy-wide effects
Bottom-up perspective	Top-down perspective

Micro and Macro Are Connected

They are **not completely separate subjects**.

The lecturer emphasizes that microeconomics and macroeconomics are interconnected.

For example:

Macro decision

Government allocates a large budget to education.



Less budget may remain for infrastructure.

Micro decision

An individual spends more time working.



Less time may be available for leisure.

Both involve **constrained choice / optimization**.

More on Economics

1. Demand

Demand = Desire + Ability to Pay + Willingness to Pay

- Desire alone ≠ demand.
- Without purchasing power, a person cannot create effective demand.

2. Microeconomics vs Macroeconomics

Micro

- Small parts of economy
- Consumers, individuals, businesses
- **Bottom-up**

Macro

- Economy as a whole
- GDP, inflation, employment, government policies

- **Top-down**

3. Three Fundamental Concepts

Scarcity + Utility + Marginality

Scarcity

Resources are limited relative to unlimited wants.

Examples:

- Land
- Labour
- Capital
- Technology
- Knowledge
- Time

Causes of Scarcity

Type	What happens	Example
Demand-induced	Supply same, demand ↑	Population growth
Supply-induced	Demand same, supply ↓	Crop failure, supply-chain disruption
Structural	Unequal access to resources	Rural vs urban facilities

4. Utility

Utility = Satisfaction obtained from consuming a good/service.

Utility can differ between people because their circumstances differ.

Example:

- Coat in cold climate → **high utility**
- Same coat in warm climate → **lower utility**

5. Marginal Utility

Marginal Utility (MU) = Additional satisfaction from consuming one additional unit.

Law of Diminishing Marginal Utility

As consumption increases, **marginal utility decreases.**

Example:

1st ice cream → highest satisfaction

2nd → lower

3rd → even lower

Overconsumption → may become negative utility

Main assumptions

- Units consumed are identical.
- Consumption occurs within a reasonable period.
- Consumer makes rational decisions.
- Units can be measured.
- Consumer's taste remains constant.
- Units aren't excessively large/small.

6. Rationality

Humans generally make decisions using **rational thinking**.

Cost-Benefit Analysis

A decision is generally made when:

Benefits > Costs

Businesses may compare factors such as:

- Price
- Product quality
- Service quality
- Brand image

7. Opportunity Cost

Opportunity Cost = Cost/value of the best alternative given up when making a choice.

Example

You have two options:

Nearby store

- Higher price
- Saves time

Supermarket

- Lower price
- Takes more time

Choosing one means giving up the benefit of the other. That sacrificed opportunity is the **opportunity cost**.

Remember

Choice → Give up alternatives → Opportunity Cost

8. Specialization / Division of Labour

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People specialize in particular tasks to earn income.

Examples:

- Farming
- Nursing
- Furniture making

This is called **division of labour**.

9. Four Economic Sectors

Sector	Main activity	Examples
Primary	Extract raw materials	Farming, fishing, mining
Secondary	Convert raw materials into products	Manufacturing
Tertiary	Provide services	IT, finance, retail, tourism
Quaternary	Knowledge/research activities	R&D, education

Basics of Demand and Supply

1. Market

A **market** is a system/place/platform where **buyers and sellers interact** to buy and sell goods or services.

It can be:

- **Organized market:** Buyers and sellers meet at a specific place/time.
Example: mandi/farmers' market.
- **Unorganized market:** Buyers and sellers may be scattered across different places/times.
Example: local ice-cream vendors.

Markets can also be classified based on **level of competition**.

2. Types of Market by Competition

Perfectly Competitive Market

Main characteristics:

1. **Many buyers and sellers**
2. Goods/services are **homogeneous or very similar**
3. **No single buyer or seller can influence the price**
4. Price is determined by **market forces**

→ Individual bargaining power is very low.

Monopoly

- **Single seller**
- Seller has **very high bargaining power**

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- The market is dominated by one player.

Oligopoly

- **Few sellers**
- Market is dominated by a small number of players.
- Sellers have relatively greater bargaining/negotiating power.

Example: Indian telecommunication industry.

Market	Sellers	Price influence
Perfect competition	Many	Very low/none individually
Monopoly	One	High
Oligopoly	Few	Relatively high

3. Demand

Definition

Demand = Quantity of a good/service that consumers are willing AND able to buy at different prices during a given period of time.

Three essential elements:

Demand = Willingness + Ability to buy + Given time period

Important

Willingness alone $\neq$ Demand

Example:

Someone wants a BMW but cannot afford it.

→ It is a **desire**, not economic demand.

Therefore:

Demand must be backed by purchasing power.

4. Law of Demand

Ceteris paribus = *other things remaining the same.*

Law

When price rises, quantity demanded falls; when price falls, quantity demanded rises, other things remaining constant.

So:

Price $\uparrow$ → Quantity Demanded $\downarrow$

Price $\downarrow$ → Quantity Demanded $\uparrow$

Example

If an ice-cream cone costs ₹1 → quantity demanded = 8

If price increases → quantity demanded decreases.

5. Demand Schedule

A **demand schedule** is a **table showing the relationship between price and quantity demanded**.

Example:

Price	Quantity Demanded
1	8
1.5	6
2	4
3	0

It states that **price is on the Y-axis** and **quantity demanded on the X-axis**.

6. Demand Curve

A **demand curve** is the **graphical representation of the relationship between price and quantity demanded**.

Shape

Downward sloping ↘

Why?

Price ↑ → Quantity Demanded ↓

Price ↓ → Quantity Demanded ↑

7. Individual Demand vs Market Demand

Individual Demand

Quantity demanded by **one consumer** at different prices during a given period.

Market Demand

Market demand = Sum of individual demands of all buyers at a given price and period.

Example

At price = ₹2:

- Person A demands = 4 cones
- Person B demands = 3 cones

Therefore:

Market demand = 4 + 3 = 7 cones

Key idea

Market Demand = Σ Individual Demands

The market demand curve shows how **total quantity demanded** changes with price.

8. Supply

Definition

Supply = Quantity of a good/service that sellers are willing AND able to offer for sale at different prices during a given period.

Three important elements:

Supply = Willingness + Ability/Capacity + Given price/time

Ability to Supply

A producer cannot supply more than their production capacity.

Example:

If a factory can produce only **10,000 units**, it cannot supply more than 10,000 units merely because the price increases.

9. Why Producers Supply More at Higher Prices

From the producer's perspective:

Price ↑ → Potential profit ↑ → Willingness to supply ↑

If price becomes very low:

Price ↓ → Profit ↓ / possible loss → Willingness to supply ↓

Therefore producers are generally motivated to supply more when prices are higher.

10. Law of Supply

Ceteris paribus = other things remaining the same.

Law

When price rises, quantity supplied rises; when price falls, quantity supplied falls.

So:

Price ↑ → Quantity Supplied ↑

Price ↓ → Quantity Supplied ↓

This is the opposite relationship to demand.

11. Supply Schedule

A **supply schedule** is a **table showing the relationship between price and quantity supplied.**

Example from the lecture:

Price	Quantity Supplied
1.5	2
3	5

12. Supply Curve

A **supply curve** is the **graphical representation of the relationship between price and quantity supplied**.

Shape

Upward sloping ↗

Why?

Price ↑ → Quantity Supplied ↑

Price ↓ → Quantity Supplied ↓

13. Individual Supply vs Market Supply

Individual Supply

Quantity supplied by **one seller/producer**.

Market Supply

Market supply = Aggregate of all individual suppliers' quantities supplied.

The market supply curve shows how **total quantity supplied** changes as price changes, assuming other factors remain constant.

Demand vs Supply — Must Know

	Demand	Supply
Concerned with	Buyers	Sellers
Key condition	Willing and able to buy	Willing and able to sell
Price ↑	Quantity demanded ↓	Quantity supplied ↑
Curve	Downward ↘	Upward ↗
Schedule	Demand schedule	Supply schedule
Market	Sum of individual demands	Sum of individual supplies

Fundamental Relationships

Demand

Price ↑ → Q^D ↓

Price ↓ → Q^D ↑

Supply

Price ↑ → Q^S ↑

Price $\downarrow \rightarrow Q^S \downarrow$

Market

Market Demand = Σ Individual Demand

Market Supply = Σ Individual Supply

Demand, Supply & Equilibrium

1. Demand Curve

A **demand curve** shows the relationship between:

- **Price** $\rightarrow$ Y-axis
- **Quantity demanded** $\rightarrow$ X-axis

Shape

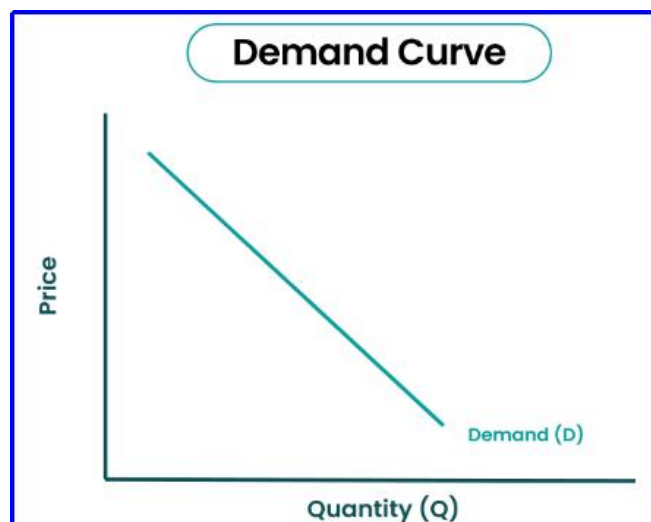
The demand curve is **downward sloping** ($\searrow$).

Price $\uparrow \rightarrow$ Quantity demanded $\downarrow$

Price $\downarrow \rightarrow$ Quantity demanded $\uparrow$

Example from the lecture:

Price	Quantity demanded
\$3	0
\$2	4
\$1	8
\$0.5	10



2. Shift in Demand Curve

A **shift in the demand curve** occurs when a variable **other than the good's own price** changes the quantity consumers demand **at a given price**.

Increase in Demand

Demand increases $\rightarrow$ curve shifts **right**

$D_1 \rightarrow D_2$

At the same price:

Quantity demanded $\uparrow$

Decrease in Demand

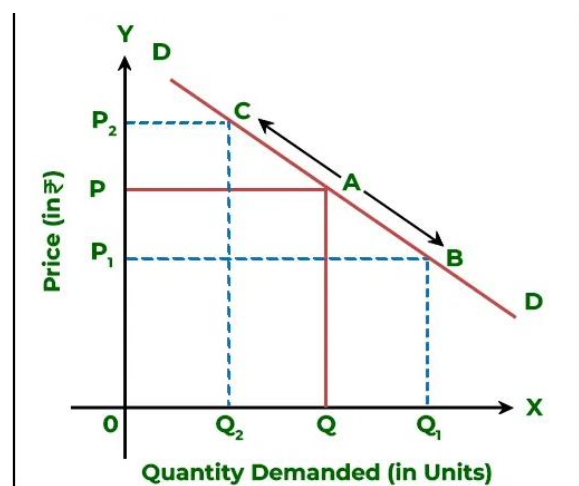
Demand decreases $\rightarrow$ curve shifts **left**

$D_2 \rightarrow D_1$

At the same price:

Quantity demanded $\downarrow$

Remember



Shift = change in demand caused by factors other than the product's own price.

3. Factors That Shift Demand

The lecture identifies several important determinants.

A. Consumer Income

For a **normal good**:

Income ↑ → Demand ↑

Income ↓ → Demand ↓

Example:

- Salaries increase
- Consumers have greater purchasing power
- Demand for consumer goods such as automobiles may increase.

Inferior Good

For an **inferior good**, demand can move in the opposite direction.

Income ↑ → Demand ↓

Income ↓ → Demand ↑

Example given in the lecture:

As income increases, a consumer may shift from **bus transportation to buying a car**.

Therefore, the analyst needs to understand whether the product is a **normal good or inferior good**.

4. Prices of Related Goods

Related goods can be:

A. Substitute Goods

Two goods that can serve as alternatives to each other.

Example:

Ice cream ↔ Frozen yogurt

If the price of ice cream falls:

Price of ice cream ↓ → Demand for ice cream ↑

Because consumers switch toward ice cream:

Demand for frozen yogurt ↓

So:

Price of substitute A ↓ → Demand for B ↓

And conversely:

Price of substitute A ↑ → Demand for B ↑

B. Complementary Goods

Goods that are generally **consumed together**.

Example given:

Ice cream + hot fudge

If the price of one complementary good falls:

Price of A ↓ → Demand for B ↑

Because the two goods are used together.

Easy memory

Substitutes → Replace each other

Complements → Consume together

5. Consumer Preferences / Tastes

Changes in **consumer preferences, tastes, trends or mindset** can shift demand.

Example:

A growing **fitness trend** can increase demand for:

- Protein shakes
- Fitness-related products/services

Therefore:

Positive change in preference → Demand ↑

Negative change in preference → Demand ↓

6. Future Expectations

Consumers' expectations about future prices can affect **current demand**.

Example:

If consumers expect **fuel prices to increase in the future**, they may purchase more petrol now.

Therefore:

Expected future price ↑ → Current demand may ↑

The important idea:

People's expectations about the future can change their purchasing decisions today.

7. Number of Buyers

An increase in the number of potential buyers increases market demand.

Example:

A company establishes a factory in an area and employs **2,000 people**.

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More people now have income and purchasing power.

Therefore:

Number of buyers $\uparrow$ $\rightarrow$ Market demand $\uparrow$

Conversely:

Number of buyers $\downarrow$ $\rightarrow$ Market demand $\downarrow$

8. Data Analyst Perspective on Demand

When forecasting demand, don't look only at the product's price.

You should consider:

- Consumer income
- Prices of substitutes
- Prices of complements
- Consumer preferences/tastes
- Future expectations
- Number of buyers
- Other relevant short-term and long-term variables

Demand forecasting requires analyzing the variables that may influence demand, not merely the product's own price.

The lecture emphasizes that these factors are **not exhaustive**.

9. Case Study: Cigarette Demand

The lecture uses cigarette smoking to demonstrate the difference between a **shift in demand** and a **movement along the demand curve**.

Situation 1: Public Awareness

Suppose the government/policymakers want to reduce smoking.

They conduct campaigns explaining the harmful effects of smoking.

Result

The **price remains unchanged**, but consumers' preferences/attitudes change.

Therefore:

Demand decreases $\rightarrow$ Demand curve shifts left

Example from lecture:

At the same price:

Original quantity = 20

After awareness campaign:

Quantity demanded $\downarrow$

Key point ★

A change in consumer preferences can shift the entire demand curve even when price does not change.

10. Movement Along the Demand Curve

Now consider a different situation.

Government imposes additional taxes on cigarettes.

Taxes increase suppliers' costs, causing the selling price to rise.

Suppose:

Price: 2 → 4

Then:

Quantity demanded: 20 → 12

But the **demand curve itself does NOT shift.**

Instead, we move from one point to another **along the same demand curve.**

Critical distinction

Change in own price → Movement along demand curve

Change in other demand determinants → Shift of demand curve

This distinction is extremely important.

11. Supply Curve

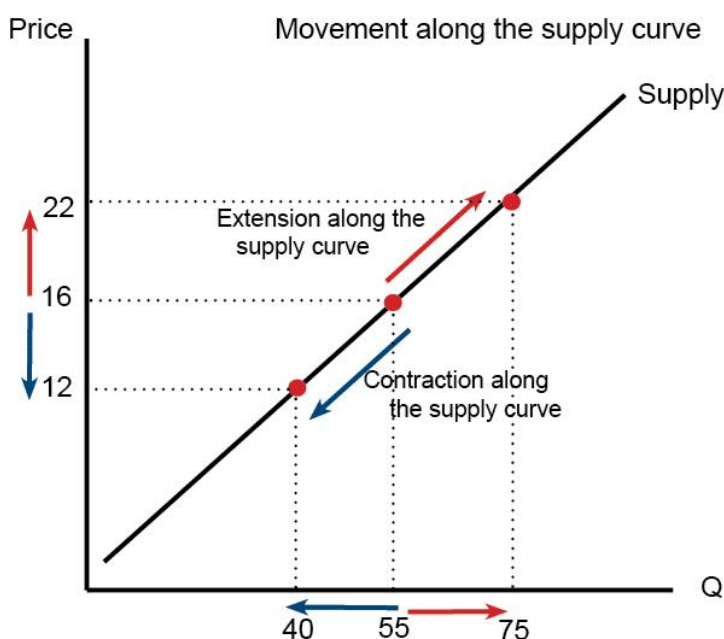
The supply curve is **upward sloping (↗)**.

From the producer's perspective:

Price ↑ → Quantity supplied ↑

Why?

Higher prices generally provide greater profit incentives.



12. Shift in Supply Curve

A supply curve shifts when factors affecting producers' willingness/ability to supply change.

Important determinants discussed in the lecture:

1. Input Prices

If input costs increase:

Input cost $\uparrow$ $\rightarrow$ Profit margin $\downarrow$ $\rightarrow$ Supply $\downarrow$

Therefore, the supply curve shifts **left**.

Example:

Raw material/labor costs increase while selling price remains unchanged.

13. Technology

Technological advancement can improve:

- Efficiency
- Mechanization
- Production capacity
- Economies of scale
- Labor productivity

This can reduce production costs.

Therefore:

Technology improvement $\rightarrow$ Cost of production $\downarrow$ $\rightarrow$ Supply $\uparrow$

So the supply curve generally shifts **right**.

14. Future Expectations of Producers

If producers expect the market price to increase in the future, they may hold/store some current production rather than sell it immediately.

Therefore:

Expected future price $\uparrow$ $\rightarrow$ Current supply $\downarrow$

The stored production can then be sold later at the expected higher price.

15. Number of Sellers

The number of sellers affects market supply.

Sellers increase

Number of sellers $\uparrow$ $\rightarrow$ Market supply $\uparrow$

Sellers decrease

Number of sellers $\downarrow$ $\rightarrow$ Market supply $\downarrow$

Example:

If several producers leave an industry:

Production capacity ↓ → Market supply ↓

16. Supply Determinants — Summary

Factor	Effect on Supply
Input prices ↑	Supply ↓
Technology improves	Supply ↑
Expected future price ↑	Current supply ↓
Number of sellers ↑	Supply ↑
Number of sellers ↓	Supply ↓

The lecture notes that these factors are **not exhaustive**; other product-specific factors can also affect supply.

17. Equilibrium Price

Now combine **demand + supply**.

The **equilibrium price** is determined at the point where:

Quantity demanded = Quantity supplied

Graphically:

Equilibrium = Intersection of Demand Curve and Supply Curve

At this point:

$$Q^D = Q^S$$

18. Equilibrium Quantity

At the equilibrium point:

Equilibrium price → price at which $Q^D = Q^S$

Equilibrium quantity → quantity at which $Q^D = Q^S$

The intersection determines **both equilibrium price and equilibrium quantity**.

19. Surplus

A **surplus** occurs when:

Market price > Equilibrium price

At this higher price:

Quantity supplied > Quantity demanded

Why?

- Producers want to supply more because price is high.
- Consumers want to buy less because price is high.

Therefore:

$$Q^D > Q^S$$

What happens?

Businesses have unsold goods.

They are likely to **reduce prices** to attract more buyers.

Price falls toward the equilibrium price.

Memory

Price above equilibrium → Surplus → Price tends to fall

20. Shortage / Excess Demand

The reverse occurs when:

Market price < Equilibrium price

At this lower price:

Quantity demanded > Quantity supplied

Why?

- Consumers want to buy more.
- Producers are less motivated to produce/sell at the low price.

Therefore:

$$Q^D > Q^S$$

What happens?

Sellers respond by **raising prices**.

Price moves toward equilibrium.

Memory

Price below equilibrium → Shortage → Price tends to rise

21. Market Adjustment

In a free-market scenario, price tends to adjust toward the level where:

$$Q^D = Q^S$$

Above equilibrium

Price > Equilibrium



Supply > Demand



Surplus



Sellers reduce price



Equilibrium

Below equilibrium

Price < Equilibrium



Demand > Supply



Shortage



Sellers raise price



Equilibrium

22. Graphical Interpretation

Demand

From the **consumer perspective**:

Lower price → Higher demand

Supply

From the **producer perspective**:

Higher price → Higher supply

Intersection

Where they meet:

Equilibrium Point

From it we obtain:

- **Equilibrium Price**
- **Equilibrium Quantity**

Movement Along Curve vs Shift of Curve

Situation	What changes?	Result
Product's own price changes	Quantity demanded	Movement along demand curve
Consumer income changes	Demand	Demand curve shifts
Taste/preferences change	Demand	Demand curve shifts
Price of substitute changes	Demand	Demand curve shifts

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Situation	What changes?	Result
Price of complement changes	Demand	Demand curve shifts
Number of buyers changes	Demand	Demand curve shifts
Expectations change	Demand	Demand curve shifts
Input price changes	Supply	Supply curve shifts
Technology changes	Supply	Supply curve shifts
Number of sellers changes	Supply	Supply curve shifts
Producer expectations change	Supply	Supply curve shifts

The golden rule:

Own price change → Movement along the curve.

Any other determinant → Shift of the curve.

Other Graphs are in the Google Drive of Student Account.

Demand Elasticity

1. What is Elasticity of Demand?

Elasticity of demand measures how responsive quantity demanded is to a change in price.

$$E_d = \frac{\% \text{ change in Quantity Demanded}}{\% \text{ change in Price}}$$

It tells us **how much demand changes when price changes.**

Simple idea

- **Large change in demand** for a small price change → **Elastic**
- **Small change in demand** for a large price change → **Inelastic**

Examples

- **Netflix subscription:** Price ↑ → many customers may cancel → **Elastic**
- **Petrol:** Price ↑ → demand may fall only slightly → **Inelastic**

2. Determinants of Demand Elasticity

The major factors discussed are:

① Availability of Substitutes

More close substitutes → **more elastic demand**

Fewer substitutes → **more inelastic demand**

Example:

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- Butter → many substitutes → relatively elastic
- Eggs → fewer close substitutes → relatively inelastic

Reason: Consumers can easily switch to another product when price increases.

② Necessity vs Luxury

Type	Elasticity
Necessity	Generally inelastic
Luxury / non-essential	Generally elastic

Examples:

- Doctor's visit → necessity → relatively inelastic
- Luxury/branded products → relatively elastic

Reason: People are less likely to stop buying necessities when prices rise.

③ Definition of the Market

Narrowly defined market → more elastic

Broadly defined market → more inelastic

Example:

- Specific **vanilla ice cream** → narrow market → more substitutes → more elastic
- **Food** as a broad category → fewer close substitutes → more inelastic
The narrower the product definition, the easier it is to find substitutes.

④ Time Horizon

Demand generally becomes **more elastic over a longer period**.

Example:

Petrol price ↑

Short run:

People cannot immediately change their cars → demand changes little → relatively inelastic.

Long run:

People can switch to EVs or other alternatives → demand changes more → more elastic.

Longer time ⇒ generally more elastic demand

3. Measuring Elasticity Quantitatively

Elasticity is measured using **percentage changes**:

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P}$$

Usually, elasticity of demand is reported using its **absolute value**, because price and quantity demanded normally move in opposite directions.

Example

Price increases by **10%**

Quantity demanded decreases by **20%**

$$E_d = \frac{20\%}{10\%} = 2$$

4. Classification of Demand Elasticity

Type	Elasticity	Meaning
Elastic	$E > 1$	% change in quantity > % change in price
Inelastic	$E < 1$	% change in quantity < % change in price
Unit Elastic	$E = 1$	% changes are equal
Perfectly Elastic	$E = \infty$	Tiny price increase → demand falls to zero
Perfectly Inelastic	$E = 0$	Quantity demanded does not change with price

5. Elastic Demand

Definition

A relatively **small change in price causes a larger change in quantity demanded**.

$$E_d > 1$$

Example: Price ↑ 10% and Quantity demanded ↓ 20%

Typical examples

- Restaurants
- Branded clothes
- Luxury/non-essential products

$$E_d = 2$$

Why?

Consumers have alternatives and can reduce or stop consumption.

6. Inelastic Demand

Definition

A relatively **large change in price causes a smaller change in quantity demanded**.

Examples

- Medicines
- Petrol
- Necessities

$$E_d < 1$$

Why?

Consumers need the product and/or have fewer substitutes.

7. Unit Elastic Demand

Here the percentage change in price and quantity demanded is **exactly equal**.

$$E_d = 1$$

Example:

Price ↓ 10% Quantity demanded ↑ 10%
So demand is **unit elastic**.

$$E_d = \frac{10\%}{10\%} = 1$$

8. Perfectly Elastic Demand

A very tiny increase in price causes quantity demanded to fall dramatically, potentially to **zero**.

Associated with:

- Highly competitive/commoditized markets
- Many close substitutes
- Buyers can easily switch sellers

$$E_d = \infty$$

Graph: Horizontal demand curve.

9. Perfectly Inelastic Demand

Quantity demanded remains **constant regardless of price**.

Example from the lecture

Life-saving insulin.

Even if price changes, the quantity required may remain essentially unchanged.

$$E_d = 0$$

Graph: Vertical demand curve.

11. Elasticity and Total Revenue ★

This is one of the **most important business applications**.

Total Revenue

$$TR = P \times Q$$

Where:

■ P = Price

■ Q = Quantity sold

When Demand is Elastic

If **price increases**:

P ↑ ⇒ Q ↓ substantially

$$E_d > 1$$

Therefore:

TR ↓

If **price decreases**, total revenue tends to increase.

Example

Price ↑ 10%, but quantity ↓ 20%.

The loss in quantity is proportionally larger than the price increase.

➡ **Revenue decreases.**

When Demand is Inelastic

If **price increases**:

P ↑ ⇒ Q ↓ slightly

Therefore:

$$E_d < 1$$

TR ↑

Because quantity doesn't fall enough to offset the higher price.

When Demand is Unit Elastic

Price changes proportionally with quantity demanded.

TR ≈ constant

$$E_d = 1$$

Elasticity of Supply

1. What is Elasticity of Supply?

Price Elasticity of Supply (PES) measures how much **quantity supplied** changes when price changes.

$$E_s = \frac{\% \text{ change in Quantity Supplied}}{\% \text{ change in Price}}$$

Supplier's perspective:

“If price changes, how easily can I change my production?”

2. Elastic vs Inelastic Supply

Business Data Management (Part 1)

Type	Meaning	Example
Elastic	Small price change → large change in supply	Clothing production
Inelastic	Price change → small change in supply	Agricultural goods in short run
Unit Elastic	% change in supply = % change in price	$E = 1$
Perfectly Elastic	Supply can change infinitely at one price	Highly competitive commodity market
Perfectly Inelastic	Supply remains fixed regardless of price	Land, rare paintings

Classification

$$E_s > 1 \Rightarrow \text{Elastic}$$

$$E_s < 1 \Rightarrow \text{Inelastic}$$

$$E_s = 1 \Rightarrow \text{Unit Elastic}$$

Example

Price of milk increases by **10%**.

Quantity supplied increases by **20%**.

$$E_s = \frac{20\%}{10\%} = 2$$

$$E_s = 2 \Rightarrow \text{Elastic Supply}$$

4. What Determines Supply Elasticity?

1. Time Horizon

More time → more elastic supply

Short run → difficult to increase production

Long run → firms can expand capacity

Example: Car manufacturer cannot double production overnight.

2. Spare Capacity

More unused capacity → **more elastic**

Example:

- Factory operating at 50% → can increase production easily → **elastic**
- Factory operating at 98% → little room to increase → **inelastic**

3. Availability of Inputs

Easy availability of:

- Raw materials
- Labour
- Machinery

→ **Higher elasticity**

Difficult to obtain inputs → **Lower elasticity**

4. Nature of Product

Perishable/unique products → generally **more inelastic supply**

Examples:

- Fresh vegetables
- Rare paintings

5. Technology

Better technology/automation → production can increase faster

Better Technology ⇒ Higher Supply Elasticity

6. Government Regulation

Heavy licensing/regulation → difficult to expand production

More Regulation ⇒ Lower Supply Elasticity

5. Elasticity Can Change Along a Supply Curve

A firm may initially have spare capacity:

Low production → easy to increase supply → more elastic

But as capacity gets full:

High production → difficult to increase supply → more inelastic

So:

Capacity Utilization ↑ ⇒ Supply Elasticity ↓

6. Technology & Supply — Wheat Example

Suppose new technology allows farmers to produce much more wheat.

Before technology

Supply = S1

After technology

Supply shifts right:

S1 → S2

Demand remains approximately unchanged.

Therefore:

More Supply → Surplus → Price Falls

If demand is relatively inelastic, quantity sold doesn't increase enough to compensate for the lower price.

Therefore:

Price ↓ ⇒ Revenue ↓

Business Data Management (Part 1)

This explains why **increasing agricultural supply can sometimes reduce farmers' total revenue.**

7. OPEC Example (Organization of Petroleum Exporting Countries)

OPEC reduces oil supply.

Short run

Oil supply is relatively inelastic because:

- Production capacity cannot change quickly
- New oil facilities take time

So:

Supply ↓ ⇒ Shortage ⇒ Price ↑

Because oil demand is also relatively inelastic in the short run:

Revenue ↑

Long run

Over time:

- New producers increase production
- Exploration increases
- Consumers use more efficient vehicles
- Energy conservation increases

Therefore:

Long-run Supply & Demand are more Elastic

So the price impact of the same supply reduction becomes smaller.

8. Economies of Scale

Economies of scale = producing more → average cost per unit falls.

Why?

Fixed costs are spread over more units.

Example:

10 units → ₹100/unit

1,000 units → ₹20/unit

Therefore:

Production ↑ ⇒ Average Cost ↓

Benefits:

- Higher profitability
- Ability to offer lower prices
- Competitive advantage
- Bulk discounts

9. Economies of Scope

Economies of scope = using the same resources to produce multiple products.

Example:

One design team works on:

- Product A
- Product B
- Product C

Instead of hiring separate teams for each.

Therefore:

Shared Resources $\Rightarrow$ Lower Total Cost

Supply Elasticity Curves

Remember:

Y-axis = Price (P)

X-axis = Quantity Supplied (Q)

Perfectly Inelastic

Vertical line:

$$E = 0$$

Quantity doesn't change even when price changes.

Example: Fixed land supply.

Inelastic

Steep upward curve:

$$0 < E < 1$$

Small change in quantity compared with price.

Unit Elastic

$$E = 1$$

Percentage change in supply = percentage change in price.

Elastic

Flatter upward curve:

$$E > 1$$

Small price change $\rightarrow$ large supply change.

Perfectly Elastic

Horizontal line:

$$E = \infty$$

Suppliers are willing to supply any quantity at one particular price.